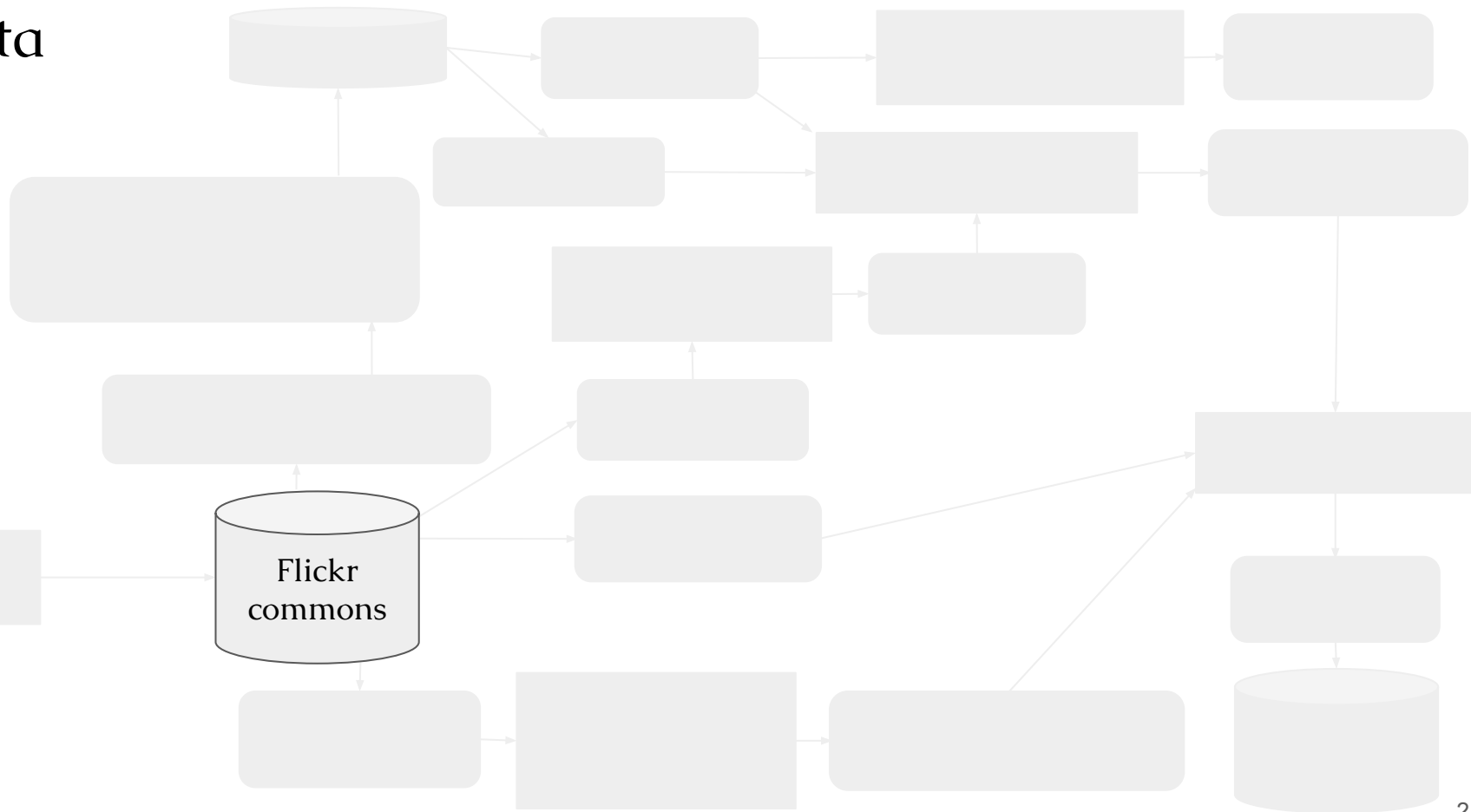


Flico: A Wilder Dataset for Dating Historical Pictures Based on Flickr Commons

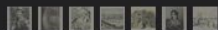
Nathanaël Lambert

Data





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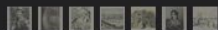
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date_taken : 2013

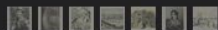
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visual : 1896

date_taken : 2013

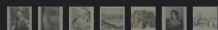
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visual : 1896

date_taken : 2013

textual 1: 1893

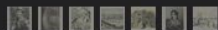
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visual : 1896

date taken : 2013
combined year: 1893

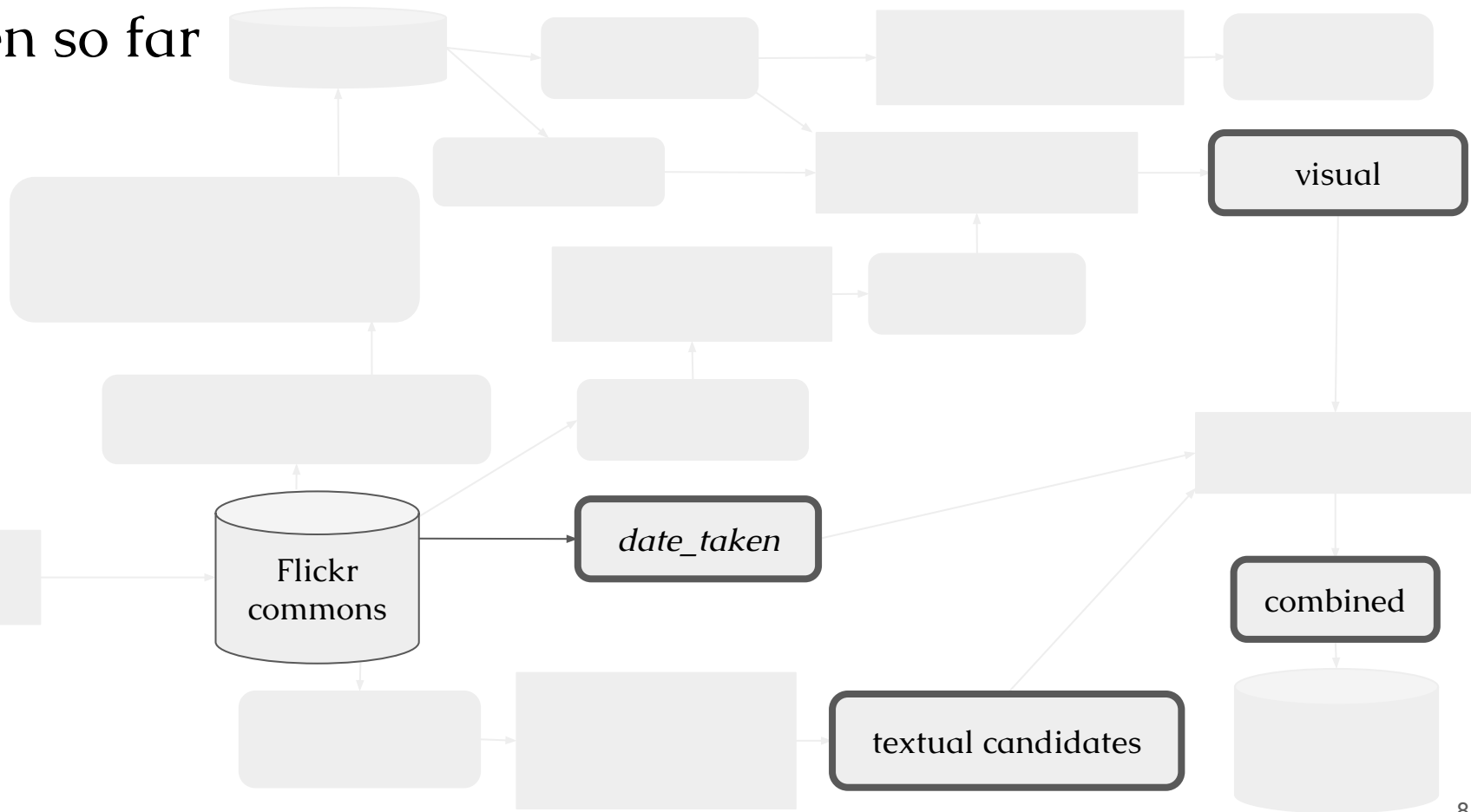
textual 1: 1893

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Uploaded on December 5, 2013

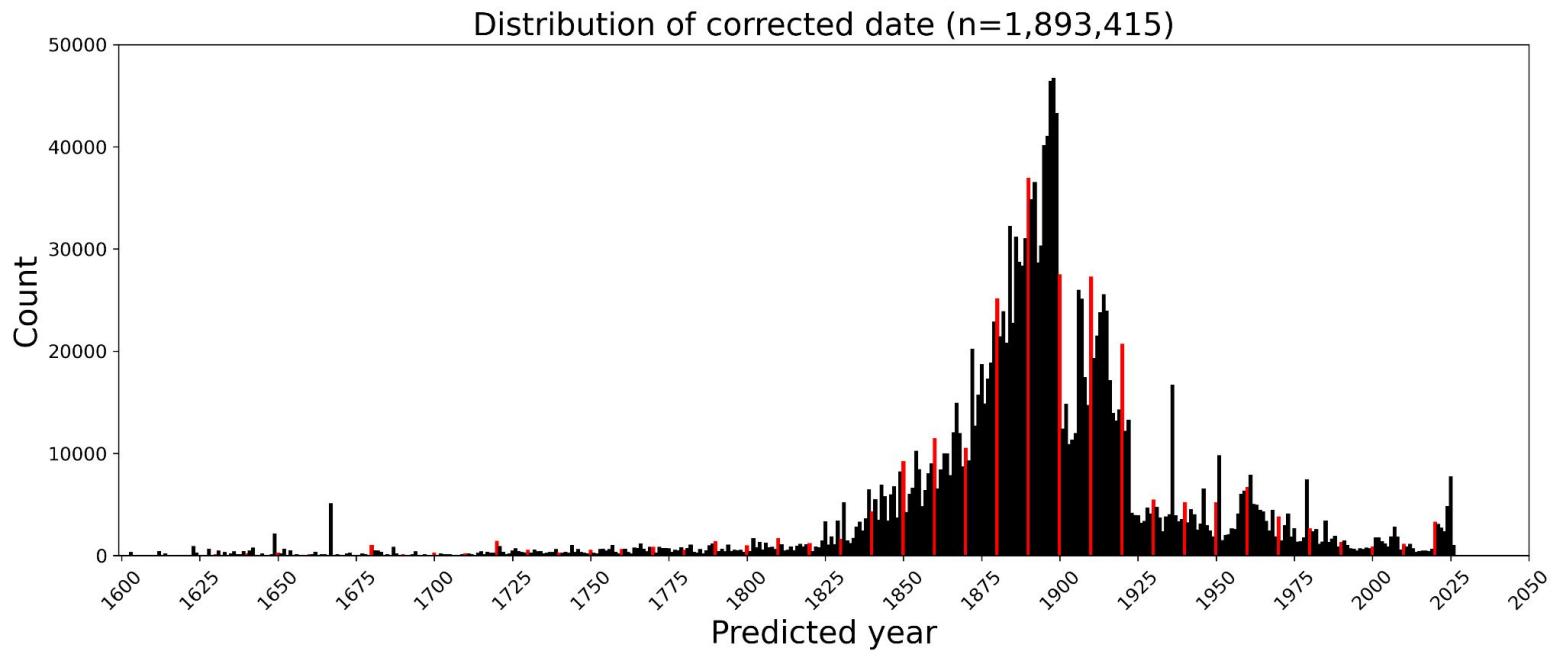
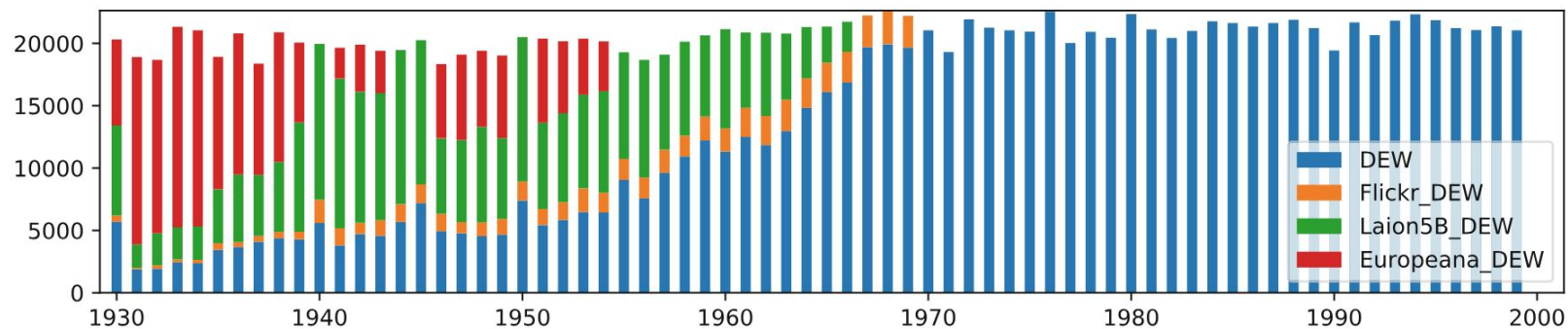
Taken on December 5, 2013

Seen so far

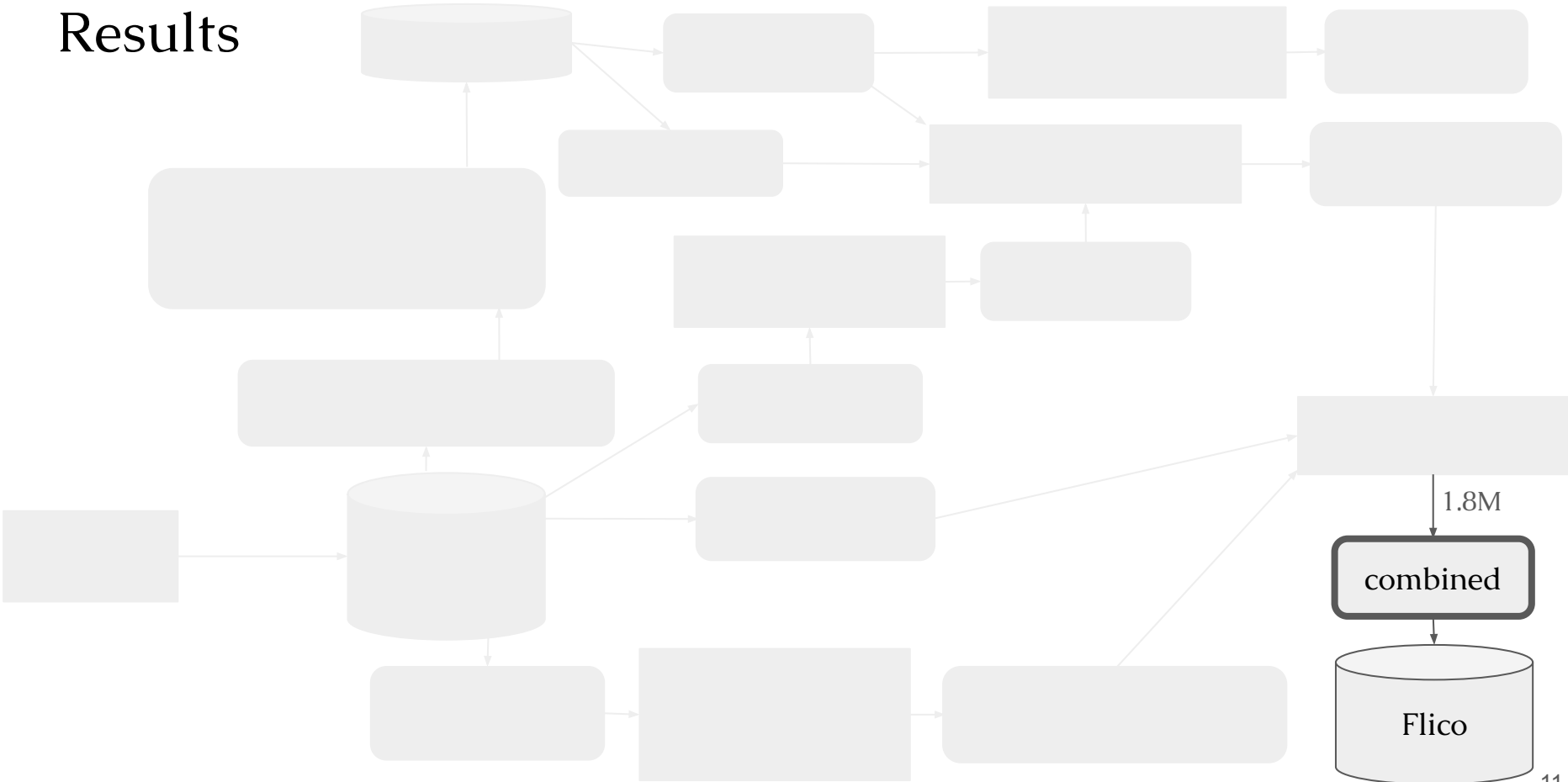


Related Work: historical pictures dating datasets

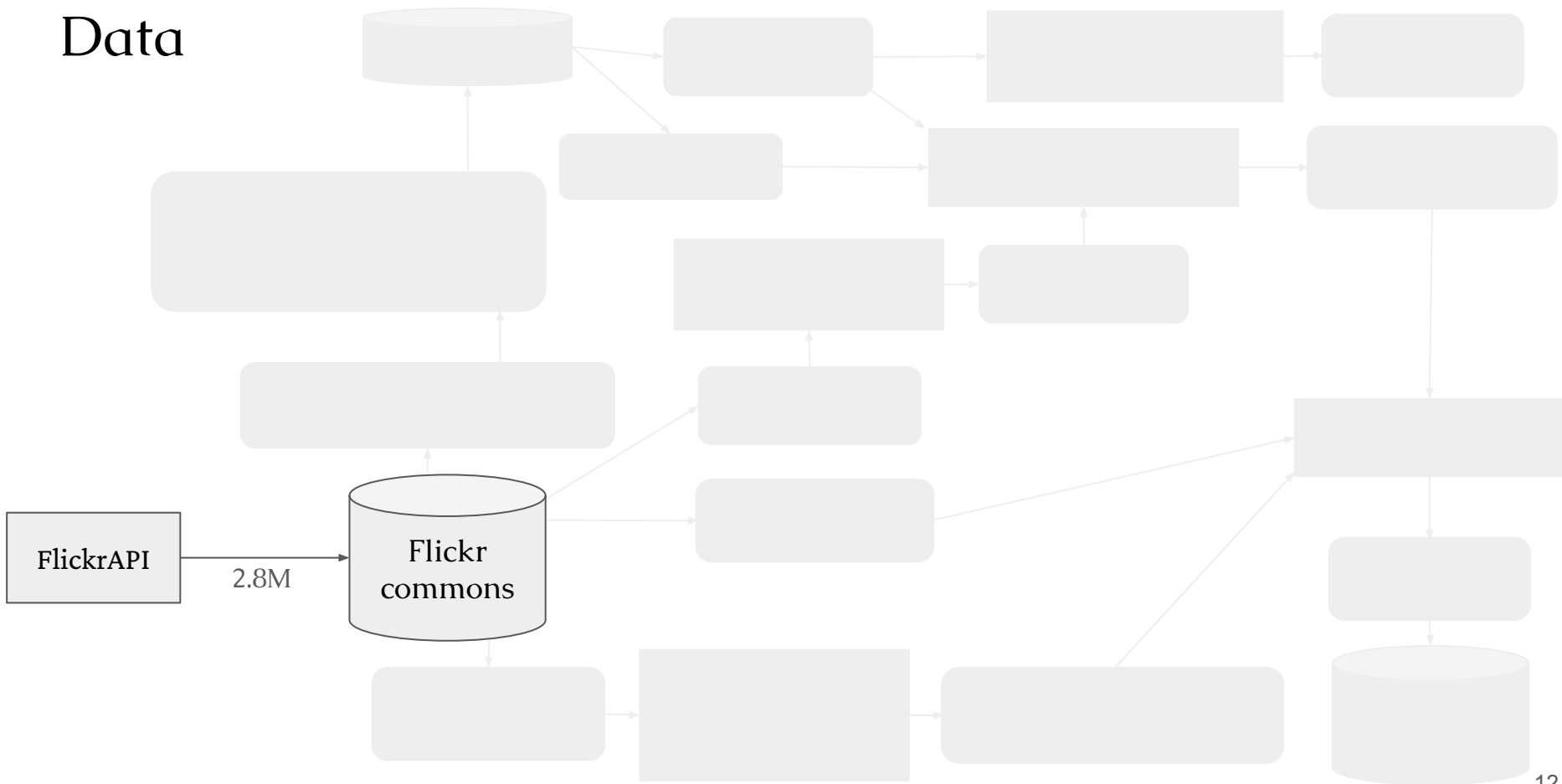
	DEW (2017)	DEW-B (2024)	Flico (2026)
Source	Flickr	DEW, Europeana, LAION-5B	Flickr Commons
Access	Largely copyrighted	Largely copyrighted	Open access
Type	Photographs	Photographs	Various types
Time frame	1930–1999	1930–1999	1000–2026
Filtering	Year as keyword	Prediction model	Prediction model
Correction	None	None	Textual extraction
Size	1,029,710	1,444,587	1,893,415



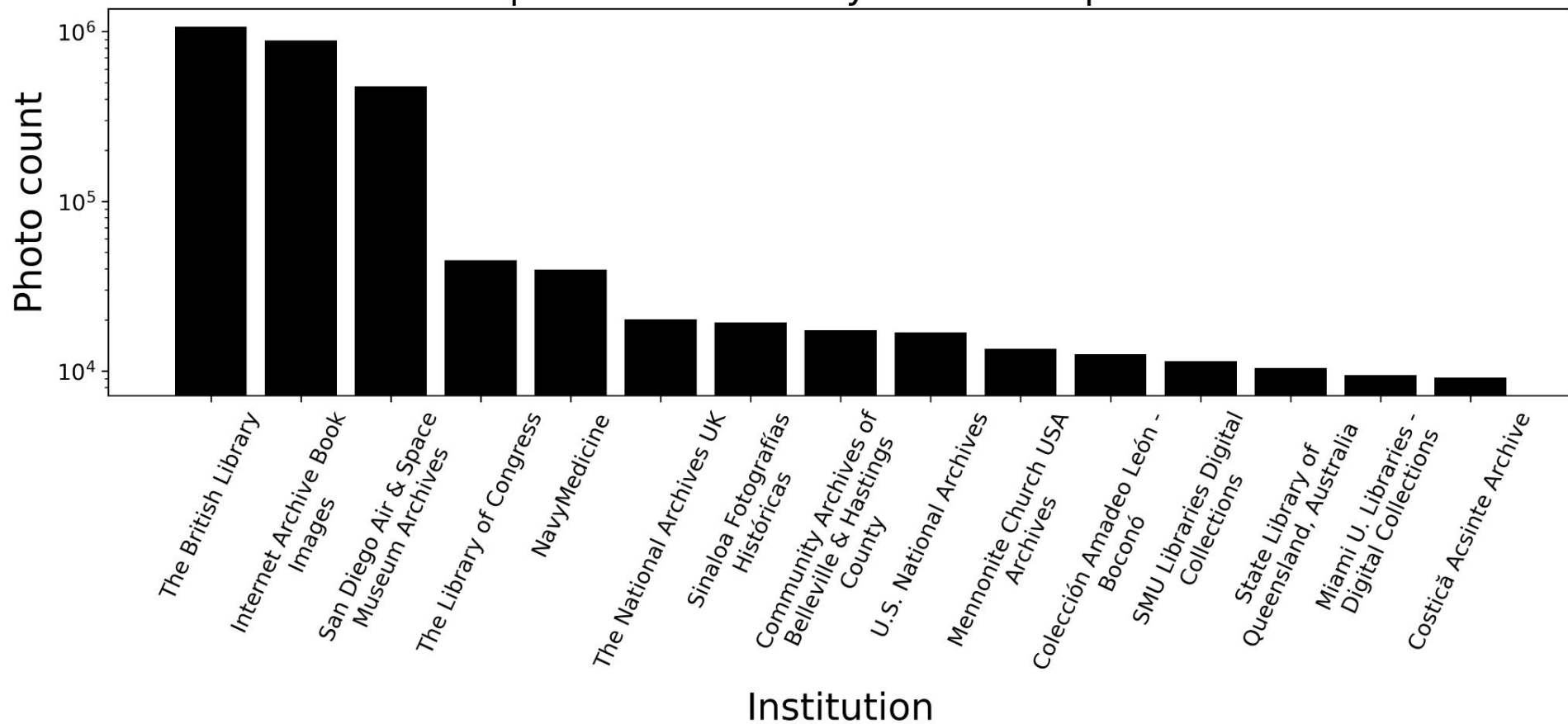
Results



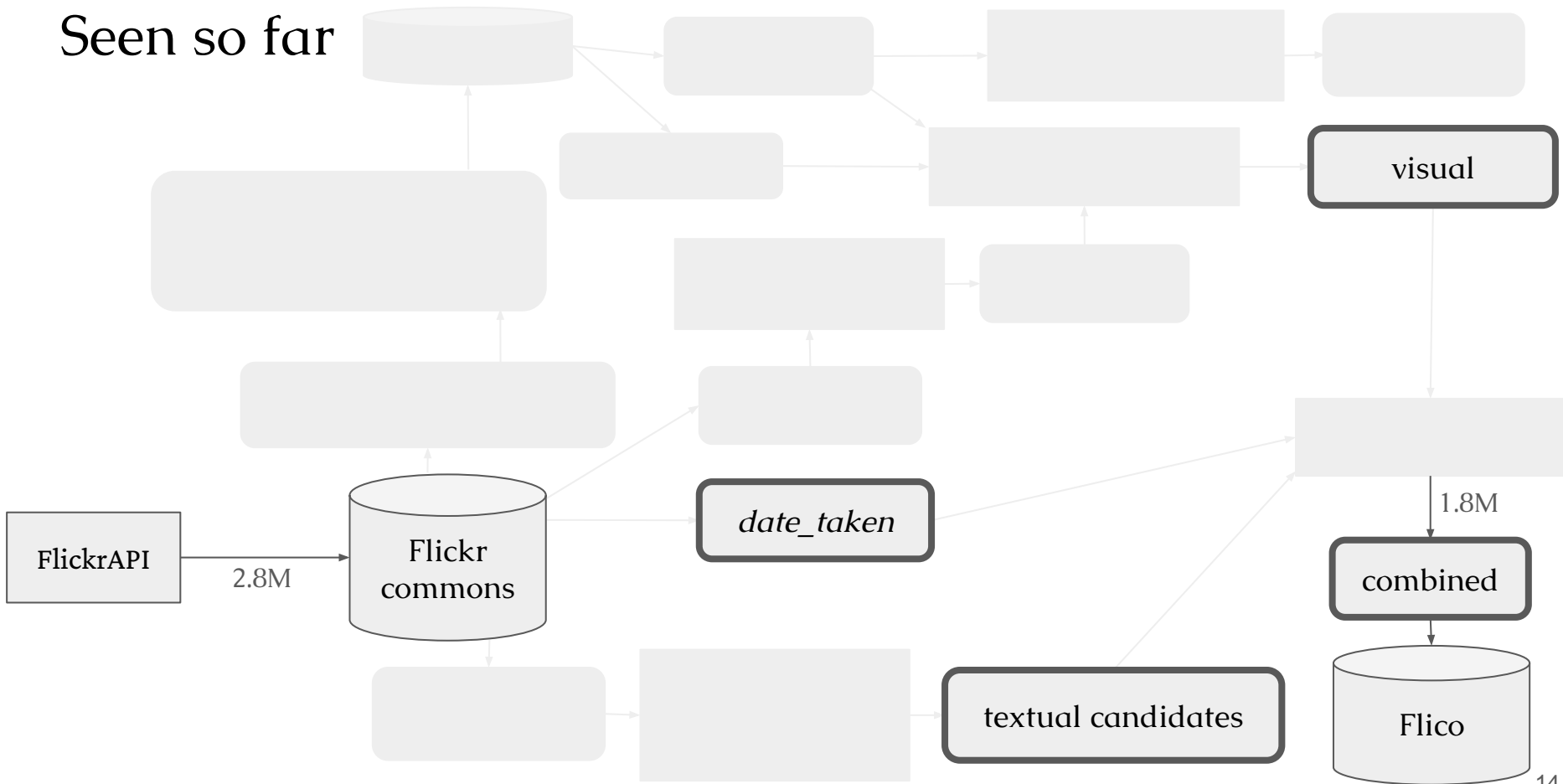
Data



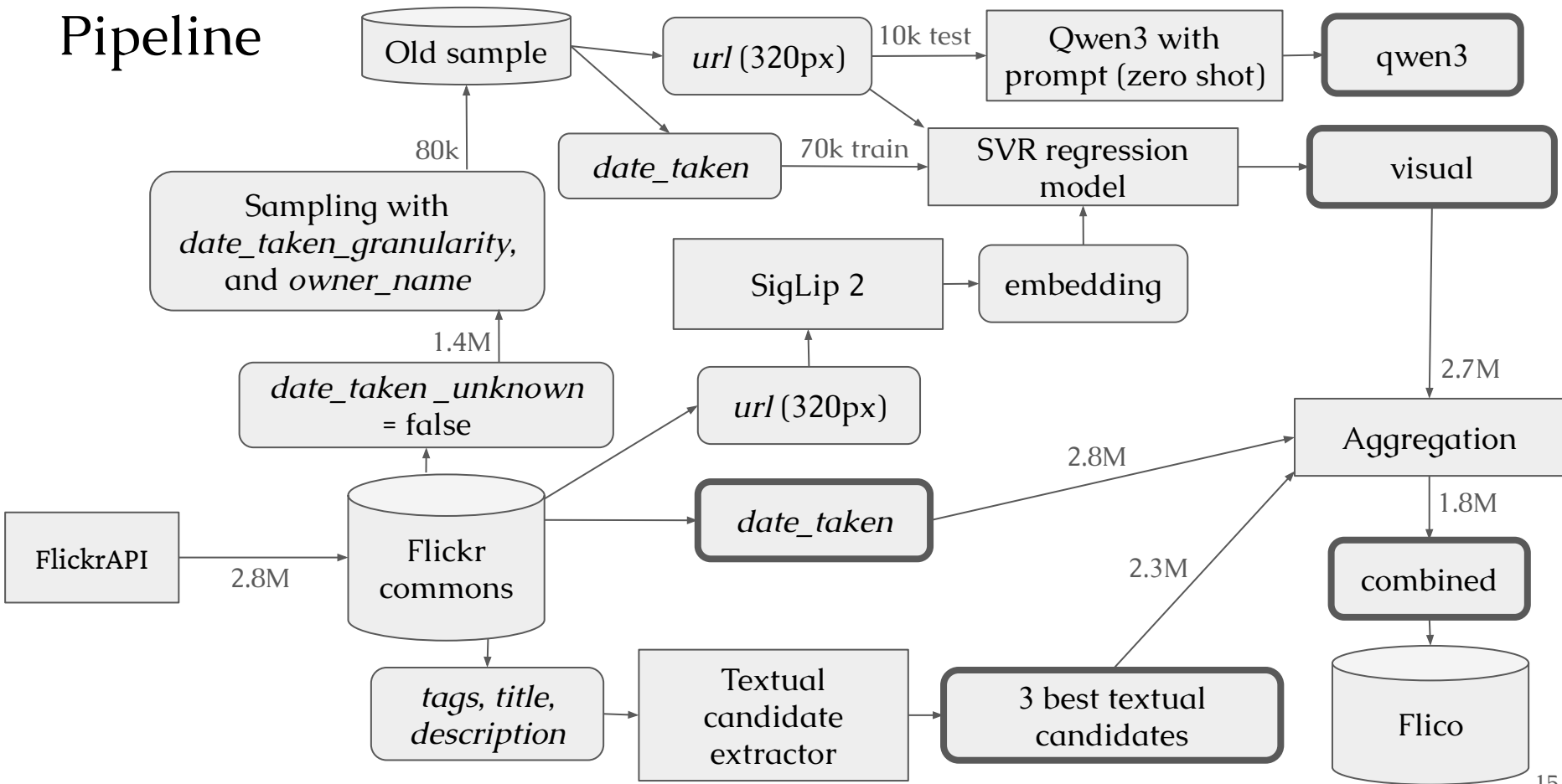
Top 15 institutions by number of photos



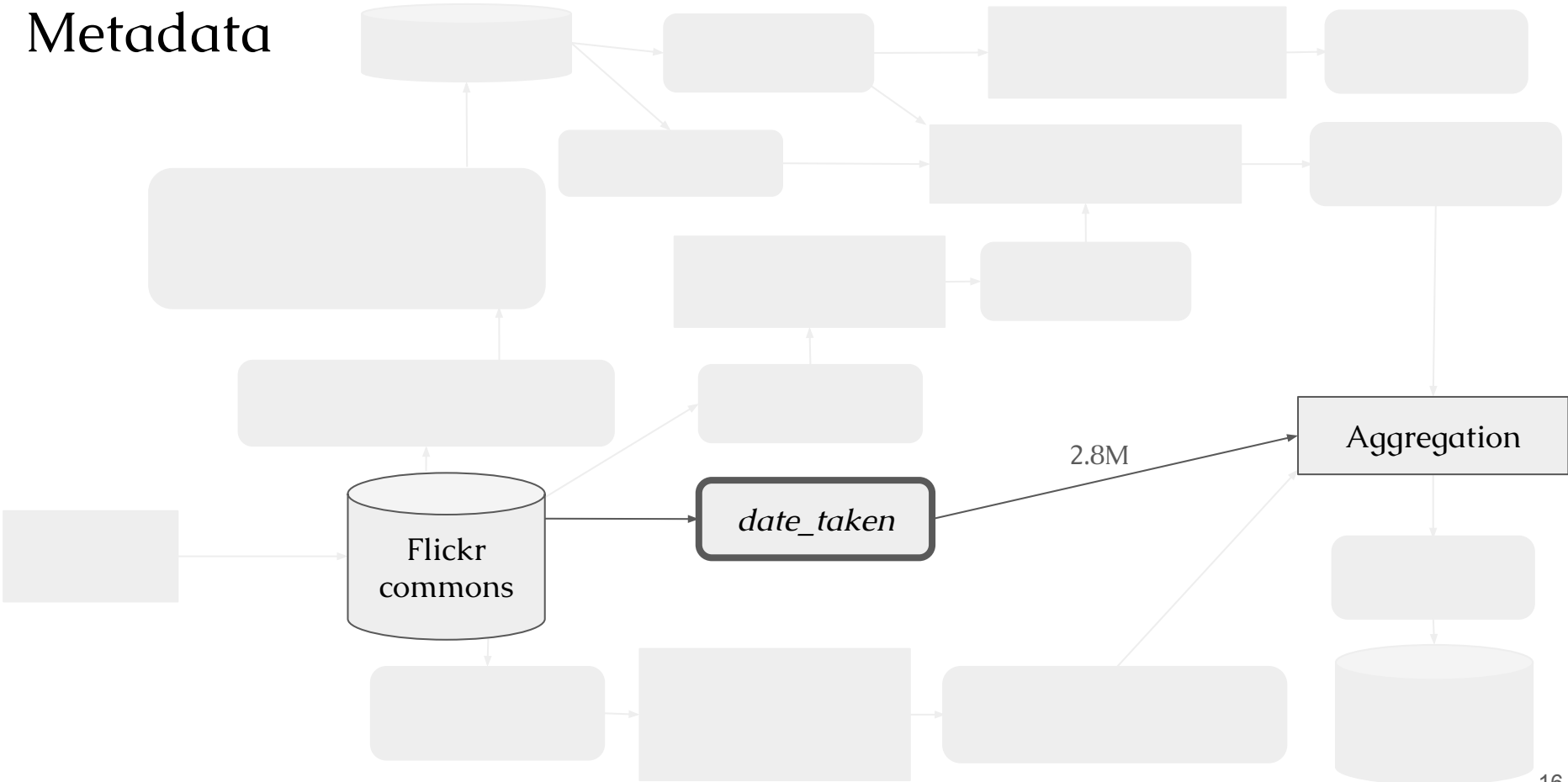
Seen so far



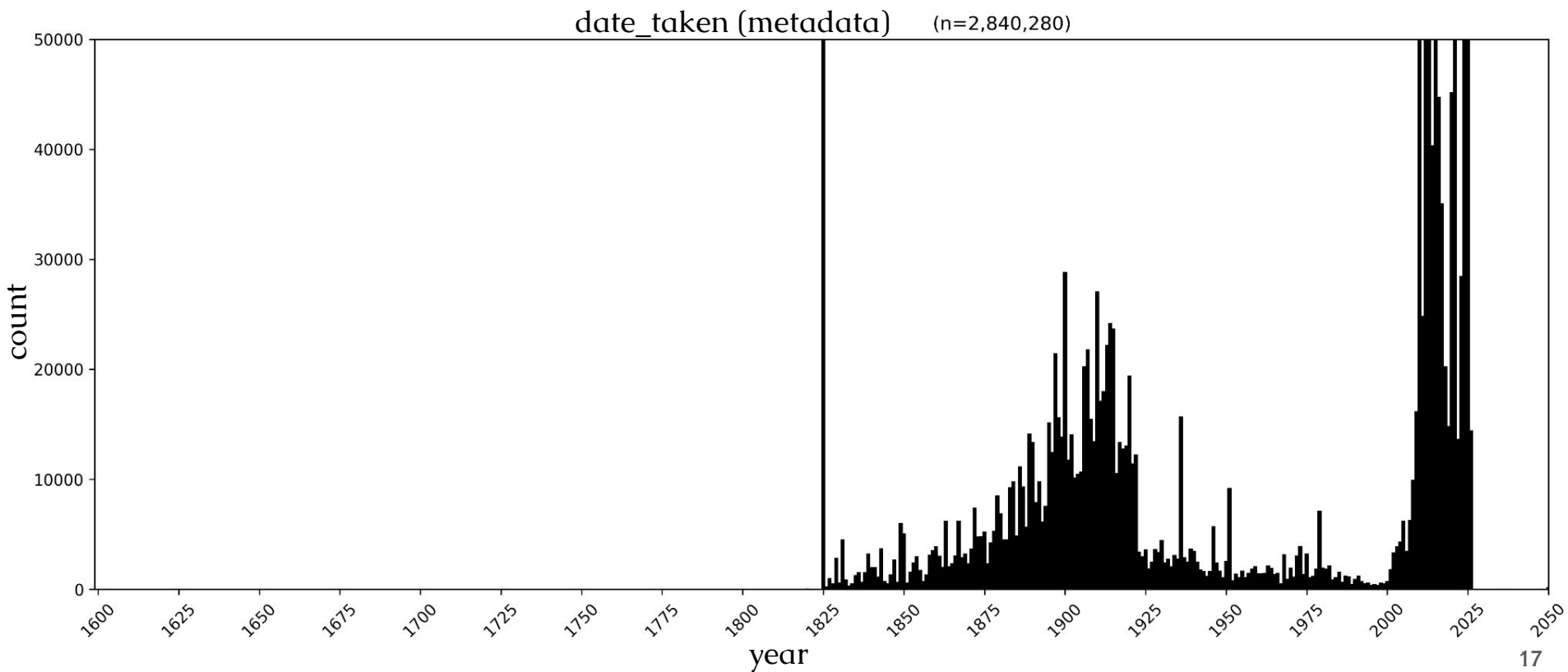
Pipeline



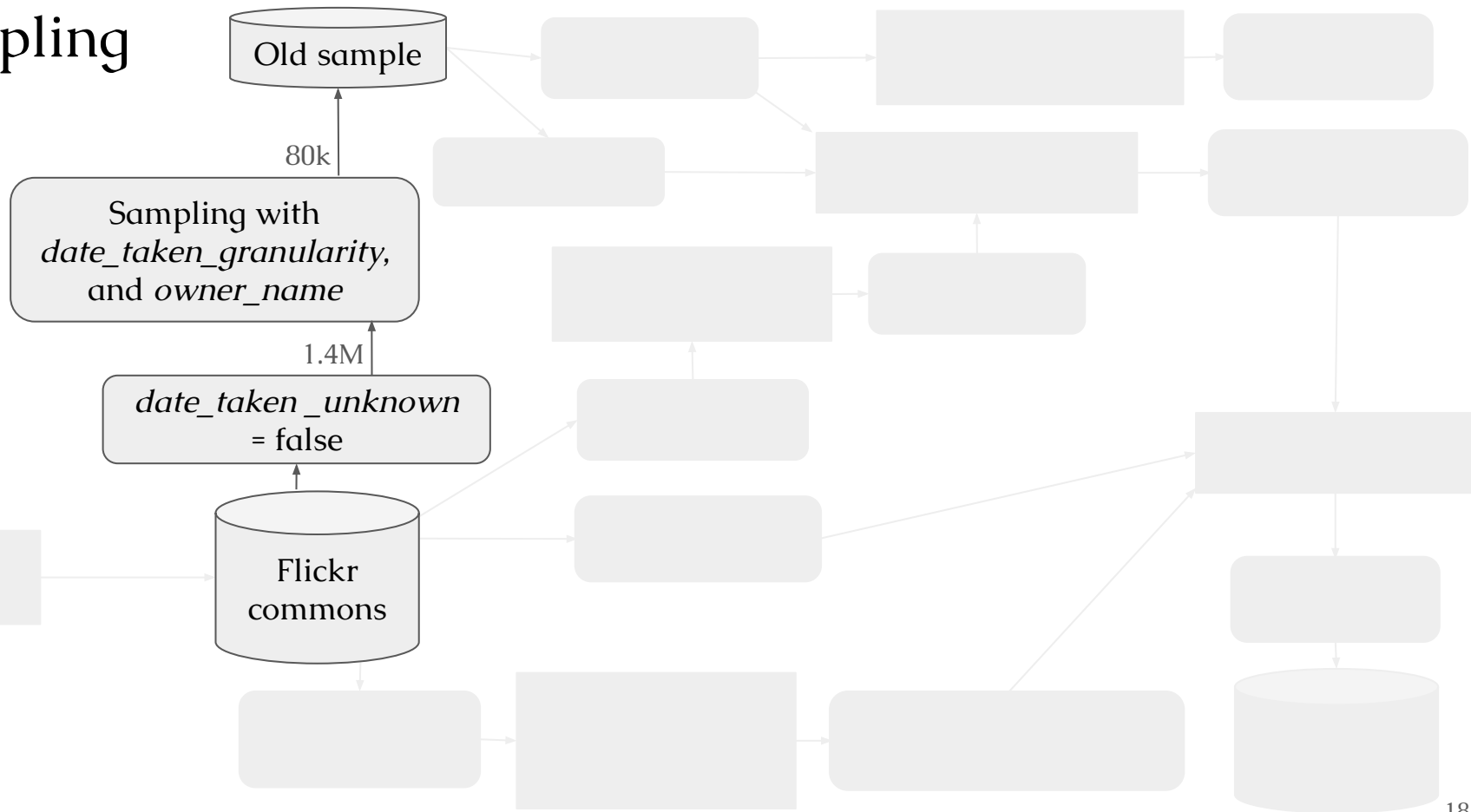
Metadata



Distribution of date_taken per year

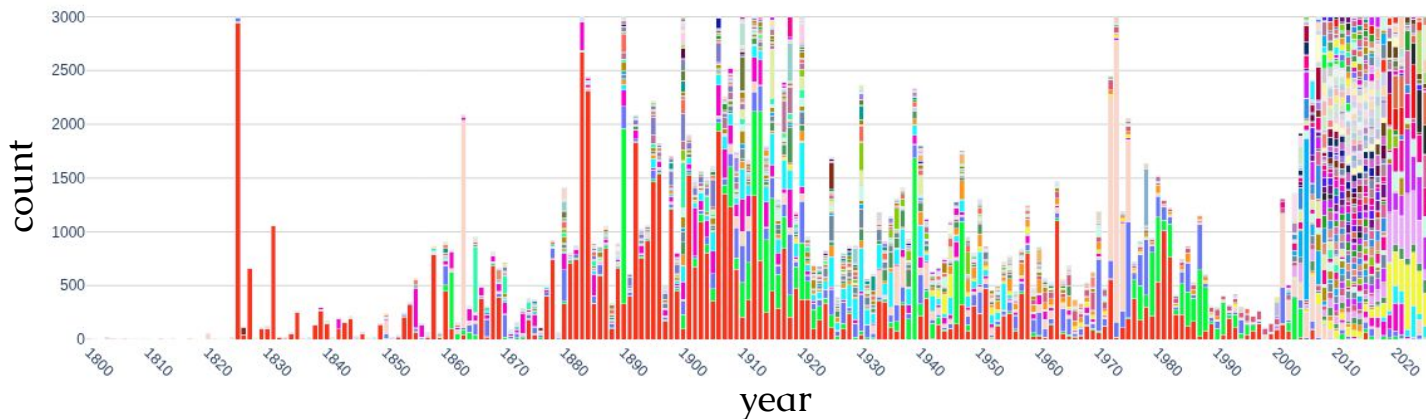


Sampling



Pictures by Year by Institutions

Before sampling:

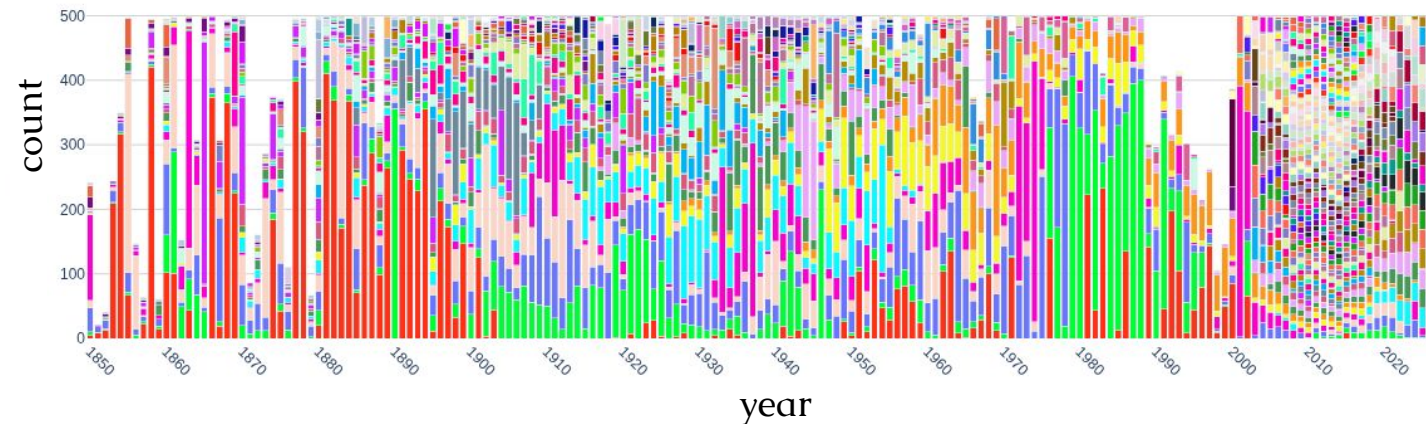


Owner Name

- Internet Archive Book Images
- The Library of Congress
- Community Archives of Belleville & Hastings County
- U.S. National Archives
- SMU Libraries Digital Collections
- State Library of Queensland, Australia
- NavyMedicine
- State Library and Archives of Florida
- State Library of NSW
- Archives of Medicine - NLM - NIH
- NASA on The Commons
- Mennonite Church USA Archives
- Australian National Maritime Museum on The Commons

Pictures by Year by Institutions

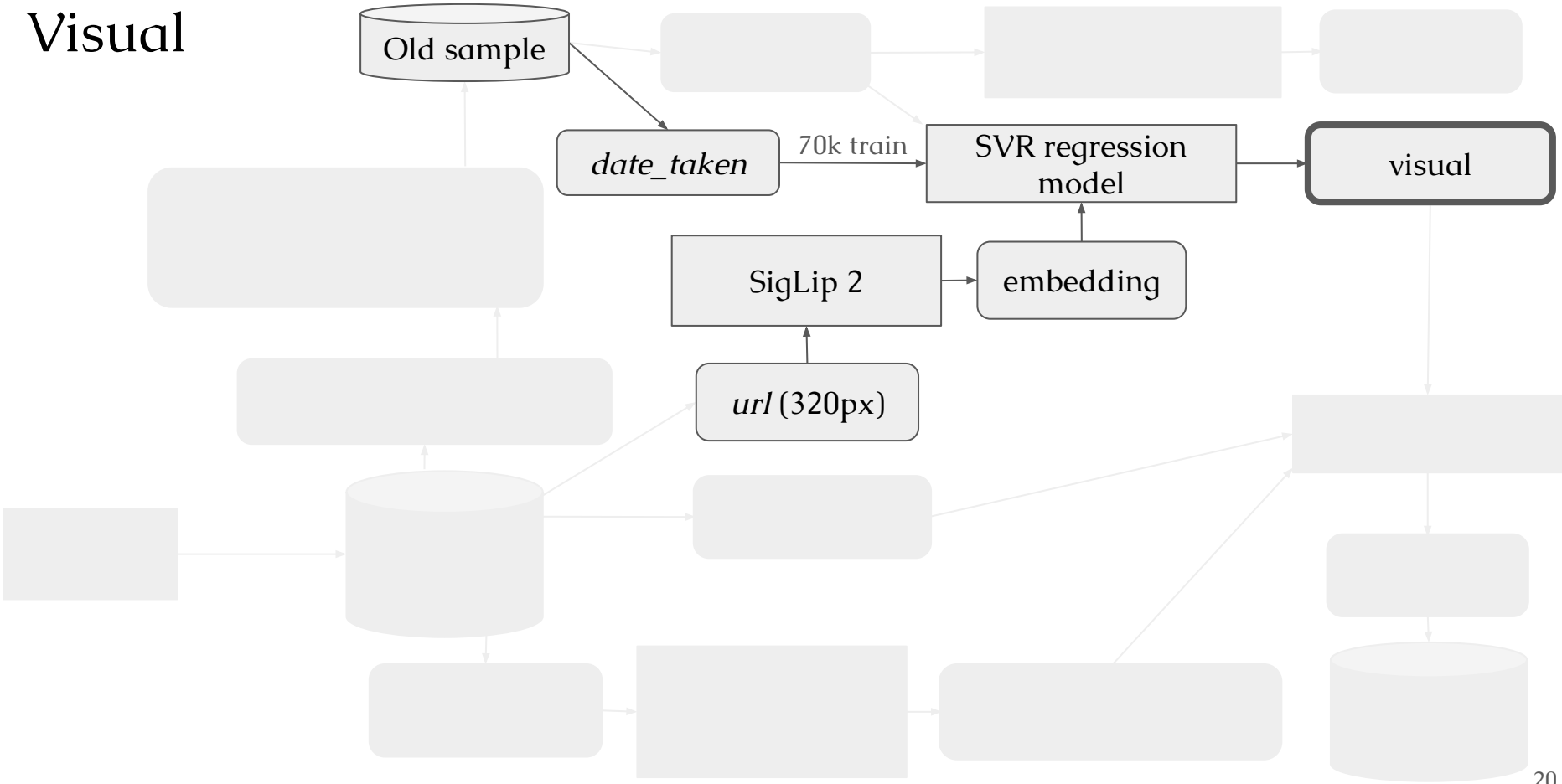
After sampling:



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- Internet Archive Book Images
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- SMU Libraries Digital Collections
- U.S. National Archives
- State Library of Queensland, Australia
- State Library and Archives of Florida
- NASA on The Commons
- State Library of NSW
- NavyMedicine
- National Library of Ireland on The Commons
- Cornell University Library
- bibliothequedetoulouse

Visual

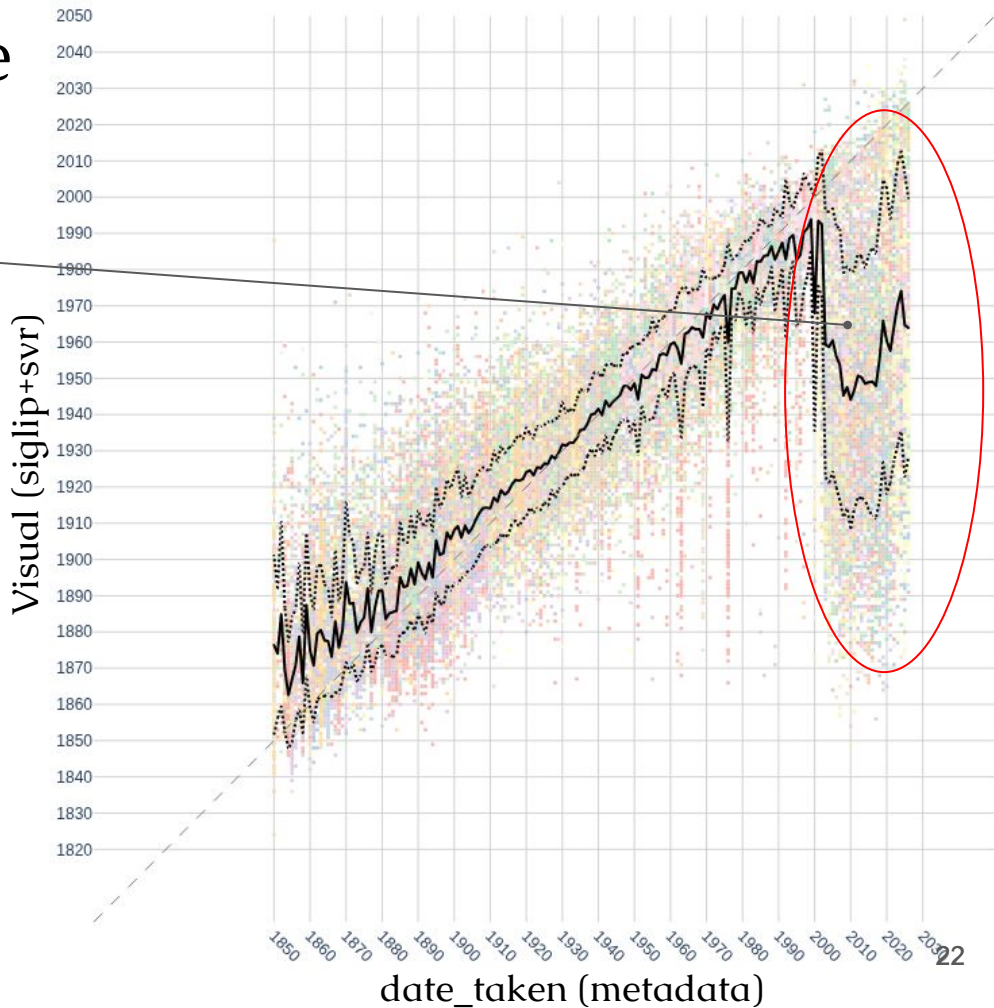


Visual models performances

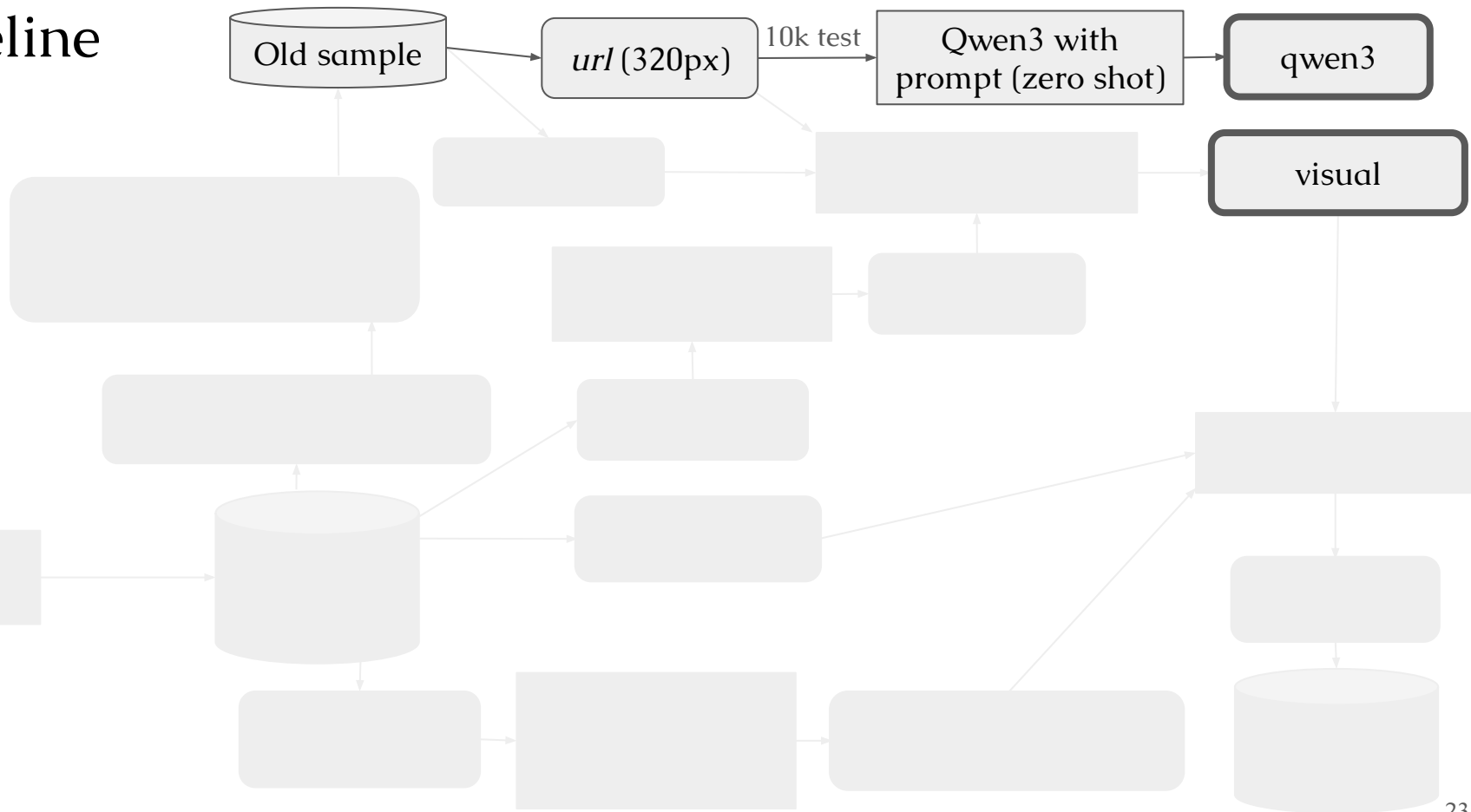
Model	Parameter	Train MAE	Test MAE
Linear Regression	—	24.98	25.45
Ridge Regression	$\alpha = 10$	24.92	25.40
	$\alpha = 2$	24.92	25.42
	$\alpha = 1$	24.92	25.41
SVR	$C = 10$	19.61	21.27
	$C = 50$	15.14	19.95
	$C = 100$	12.37	19.60

The “scanned date” issue

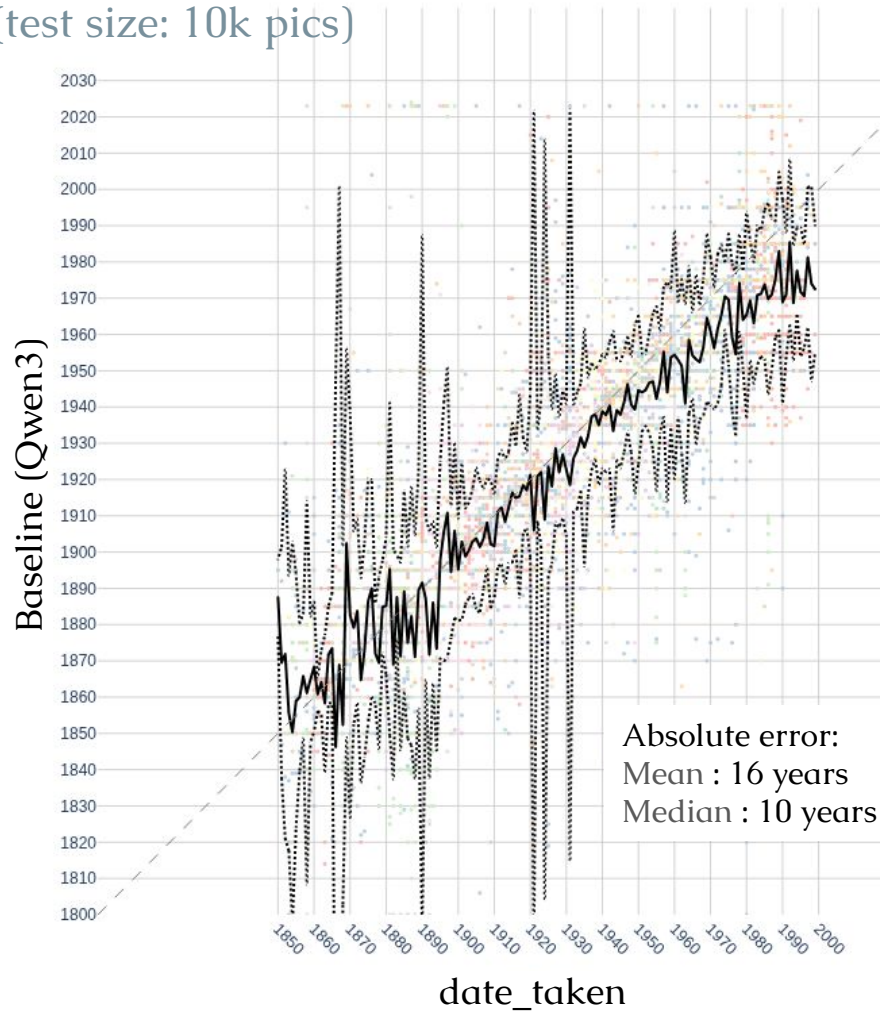
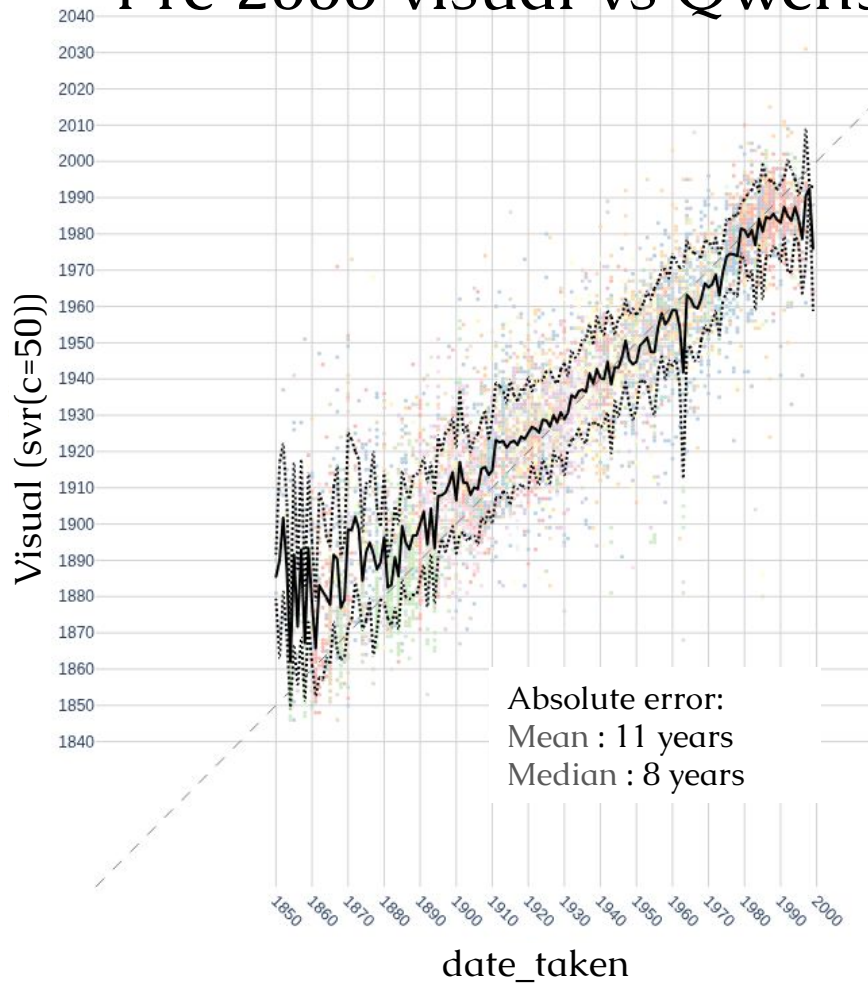
date_taken : 2010
actual : 1964



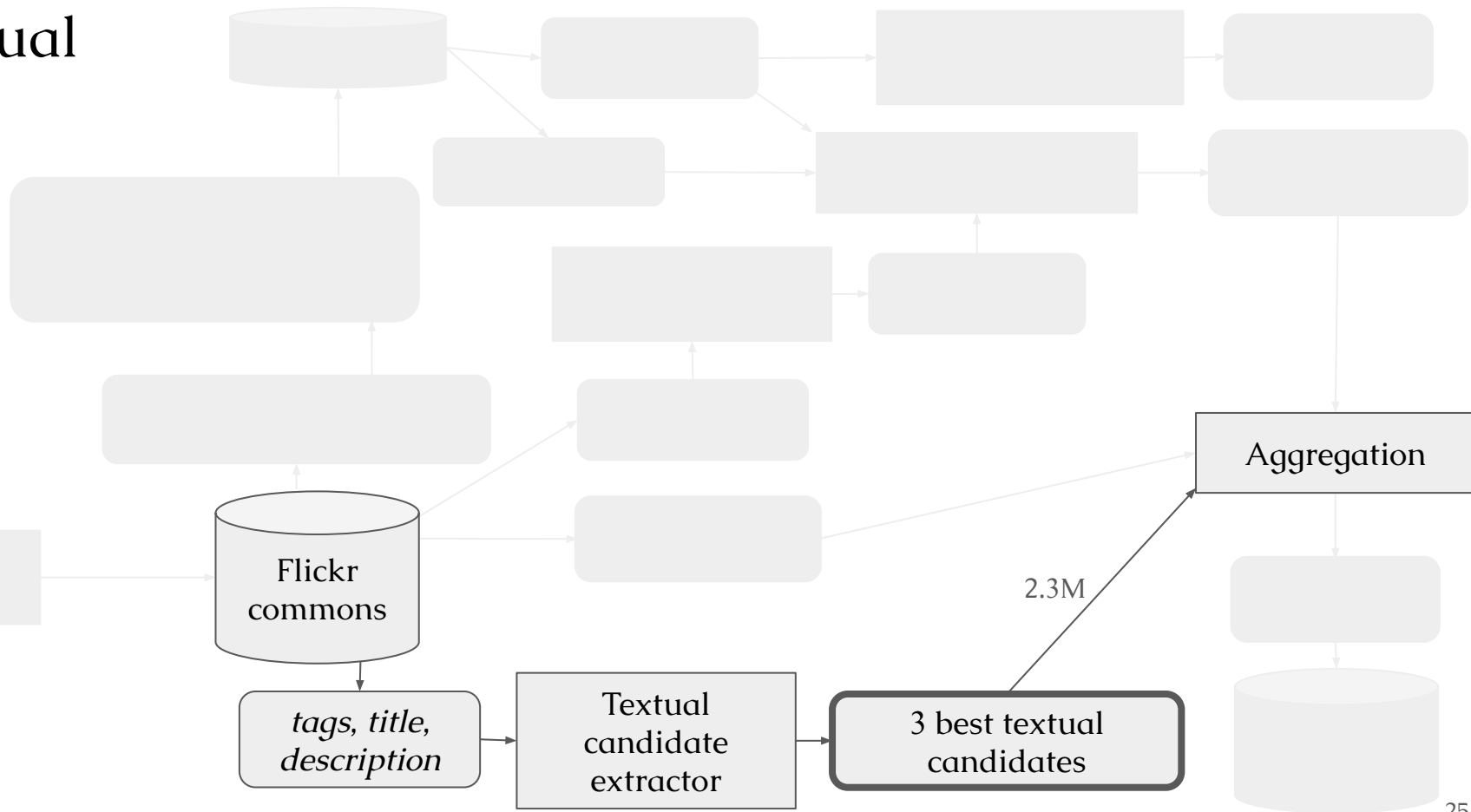
Baseline



Pre 2000 visual vs Qwen3 (test size: 10k pics)



Textual



Scoring candidates

	Pattern	Score
	There is a bracket before the candidate	+13
	There is a bracket after the candidate	+14
	There is a parenthesis before the candidate	+7
	There is a parenthesis after the candidate	+8
	There is no other date on the line	+10
	The candidate appears in the title	+5
	The candidate appears in the tags	+12
	There is an earlier date in the description	+6
	There is a later date in the description	+6
	The keyword 'date' appears on the same line as the candidate	+28
	The keyword 'year' appears on the same line as the candidate	+28
	The keyword 'produced' or 'created' appears in the same sentence as the candidate	+30
	The keyword 'circa' appears in the same sentence as the candidate	+6
	The candidate date finishes with a '0'	-7
	'no date recorded' or 'ukjent datering' appear in the description	-50
	The candidate date is older than 2030	-300
	The candidate date is a serial number like '1524-AB-03'	-80
	The keyword 'CO' or 'ref' appears in the same sentence as the candidate	-40
	The keyword 'donated' or 'transfer' appears in the same sentence as the candidate	-50
	The keyword 'number' or 'no.' appears in the same sentence as the candidate	-50
	A '\$' appears in the same sentence as the candidate	-50
	The keyword 'street' or 'road' appears in the same sentence as the candidate	-40

Example

Title: How did this cat make the news in 1924? (LOC)

Description: Find out in our recent Picture This blog post: Coolidge's Cat: Out of the Bag!. — — — "Tige" the White House cat and pet of Mrs. Coolidge has been returned. 1924 March 25. 1 photographic print Summary: Benj. Fink, half-length portrait, standing, facing front, holding "Tige" the White House cat. Notes: • Caption continues: "Benj. Fink, guard at the Navy Dept. found Tige promenading around the Navy Bldg. and immediately returned him to the White House. "Tige's" disappearance was broadcasted by Wash. radio stations." • In album: 1 March to 30 April 1924, v. 2, Herbert E. French, National Photo Company, p. 25, no. 29633. For more about the album, see: www.loc.gov/pictures/collection/coll/item/2004668683/ • Forms part of: National Photo Company Collection (Library of Congress). • Original glass negative may be available: LC-F8-29633 Subjects: • Coolidge, Grace Goodhue, 1879-1957-Animals & pets. • ets-1920-1930. • Cats-Washington (D.C.)- 10https://www.flickr.com/photos/library_of_congress/43387018050/in/datetaken/1920-1930. Format: • Photographic prints-1920-1930. • Portrait photographs-1920-1930. Rights Info: No known restrictions on publication. Repository: Library of Congress, Prints and Photographs Division, Washington, D.C. 20540 USA, hdl.loc.gov/loc.pnp/pp.print Higher resolution image is available (Persistent URL): hdl.loc.gov/loc.pnp/cph.3c31880

Tags: Library of Congress, Tige, Cat, Presidentia Pets, Pet



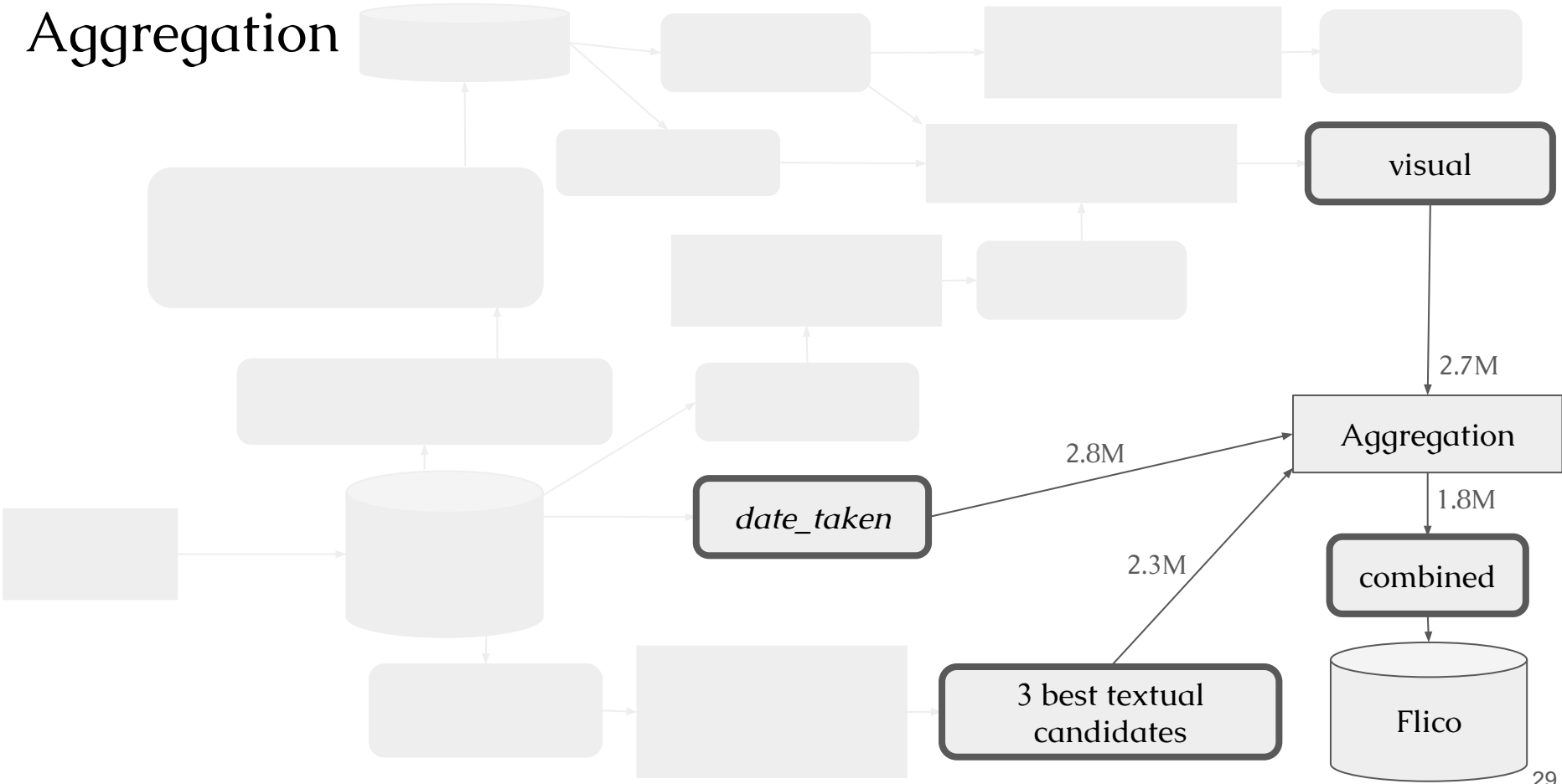
Candidate	Score	Source	Context
1924	56.99	title	cat make the news in 1924? (LOC)
1924	51.99	description	1924 March 25
1924	41.99	description	In album: 1 March to 30 April 1924
1879	34.62	description	1879 –1957–Animals & pets
1957	35.32	description	1879– 1957 –Animals & pets
1920	34.98	description	Pets– 1920 –1930
1930	34.98	description	Pets–1920– 1930
1920	34.98	description	Cats–Washington (D.C.)– 1920 –1930
1930	34.98	description	Cats–Washington (D.C.)–1920– 1930
1920	34.98	description	Photographic prints– 1920 –1930
1930	34.98	description	Photographic prints–1920– 1930
1920	34.98	description	Portrait photographs– 1920 –1930
1930	34.98	description	Portrait photographs–1920– 1930

Table 5: Score per candidate

Year	Probability
1924	0.853
1920	0.748
1930	0.748
1957	0.718
1879	0.715

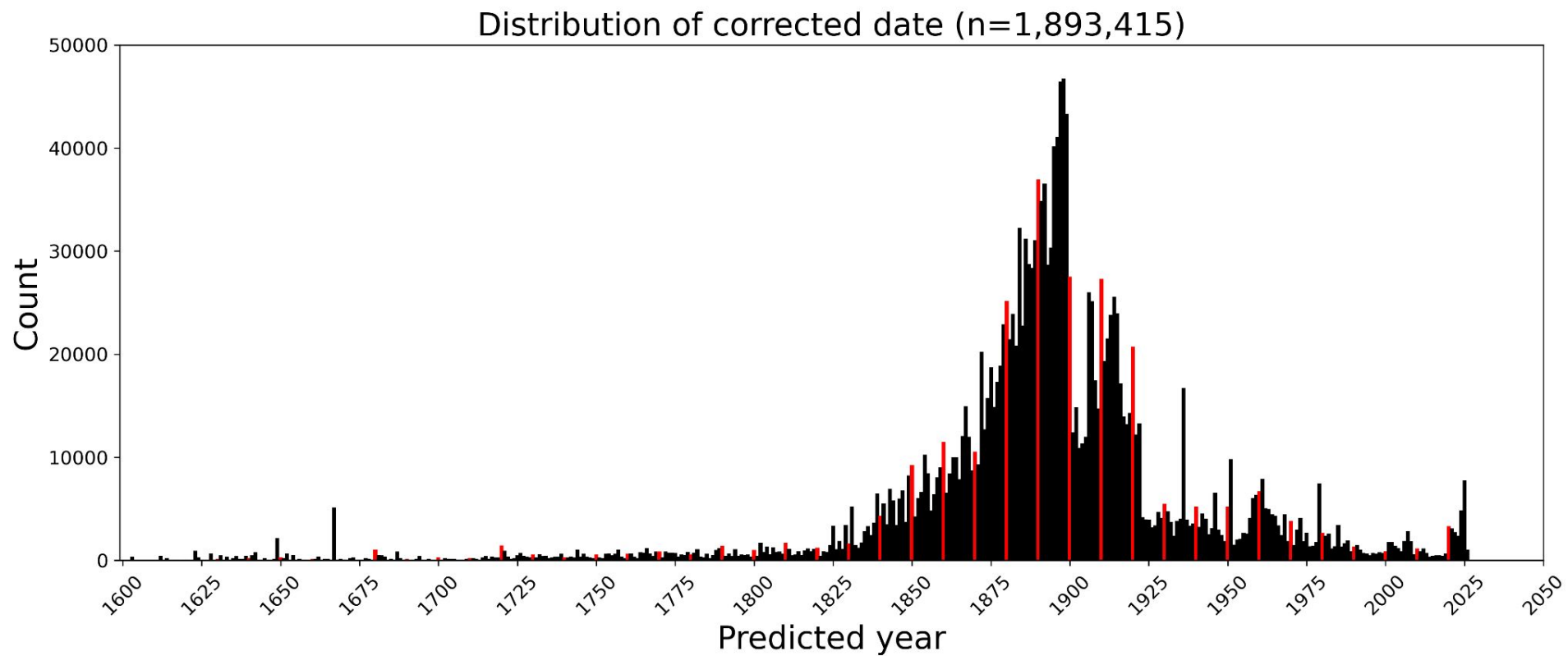
Table 6: Probability score per year

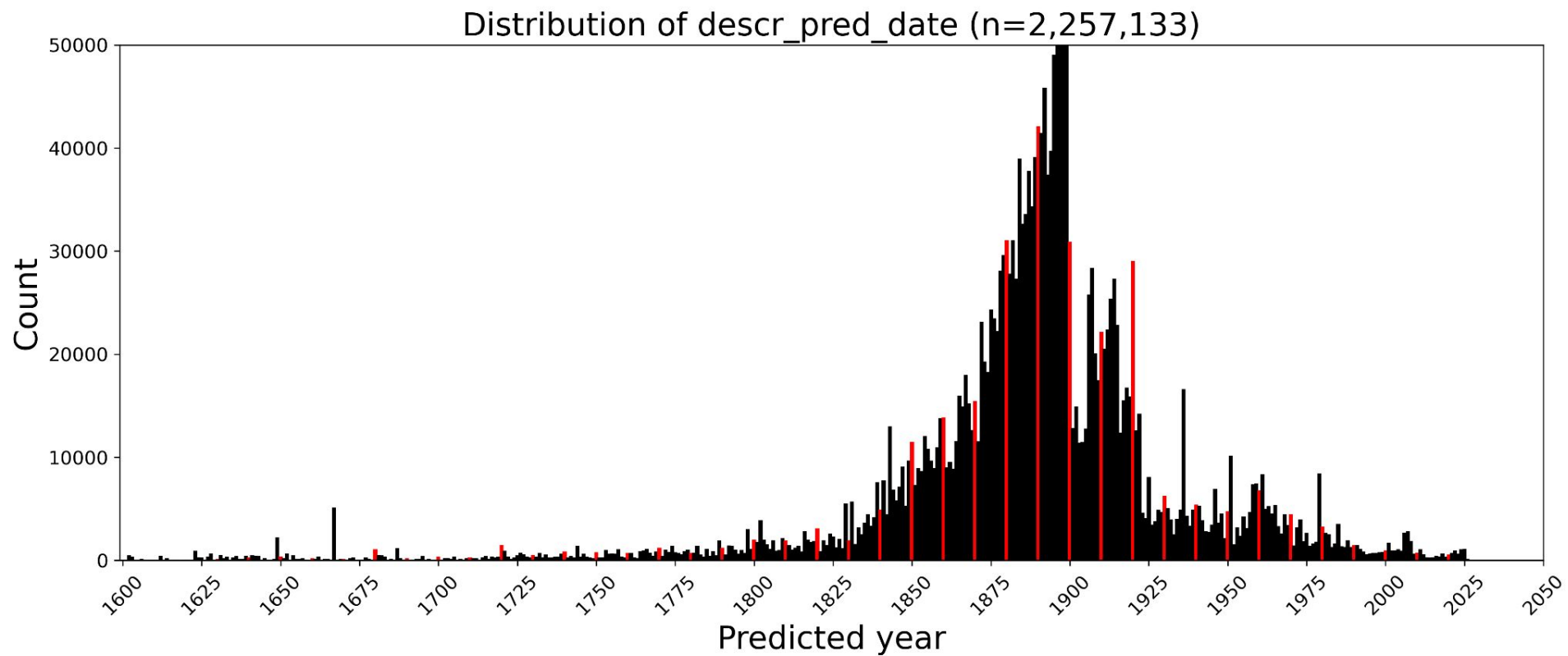
Aggregation

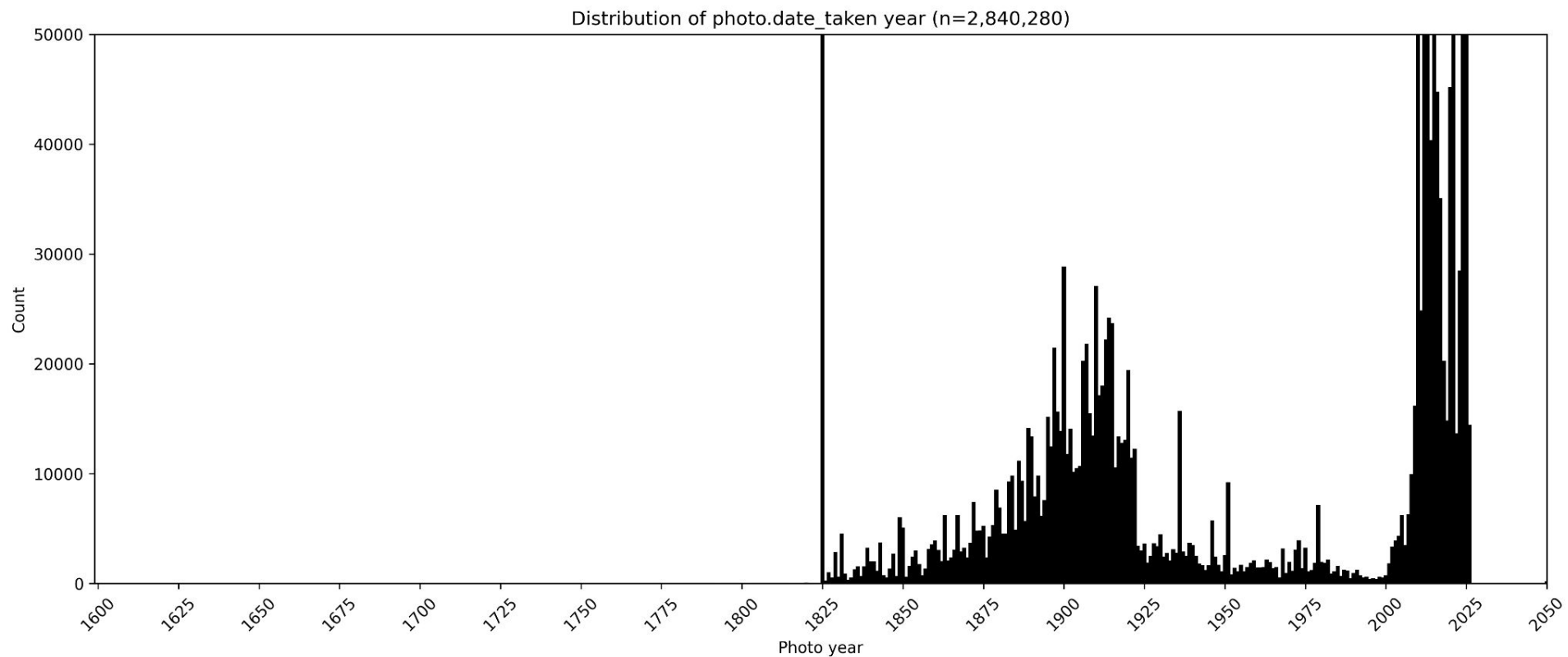


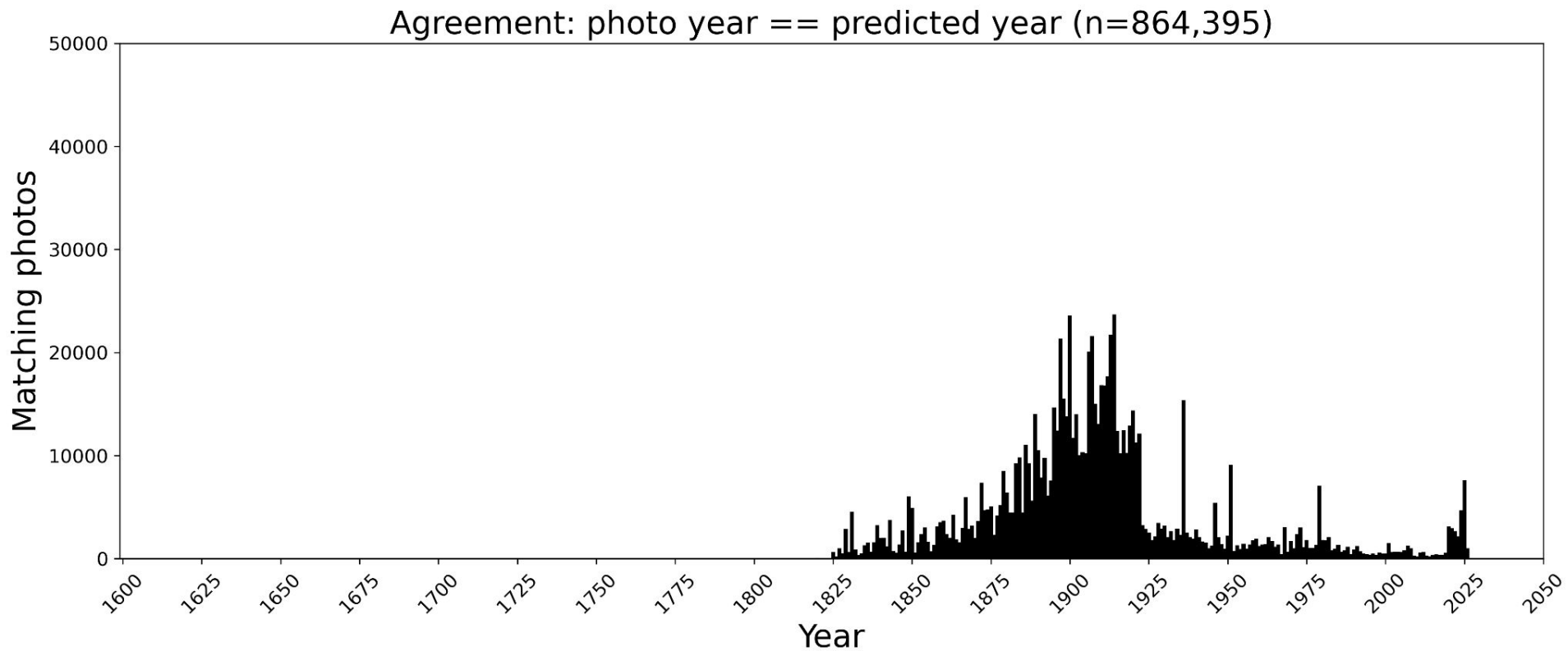
Mean Absolute Error (on 600 hand-annotated pictures)

Period	Aggregated	Metadata	Text	Visual
<1800	0.21	259.23	1.77	230.96
1800–1849	0.08	87.45	0.01	82.95
1850–1899	1.07	70.38	14.88	32.98
1900–1949	1.25	46.23	23.35	15.41
1950–1999	0.44	26.60	25.07	16.35
≥ 2000	5.13	5.46	6.25	13.78

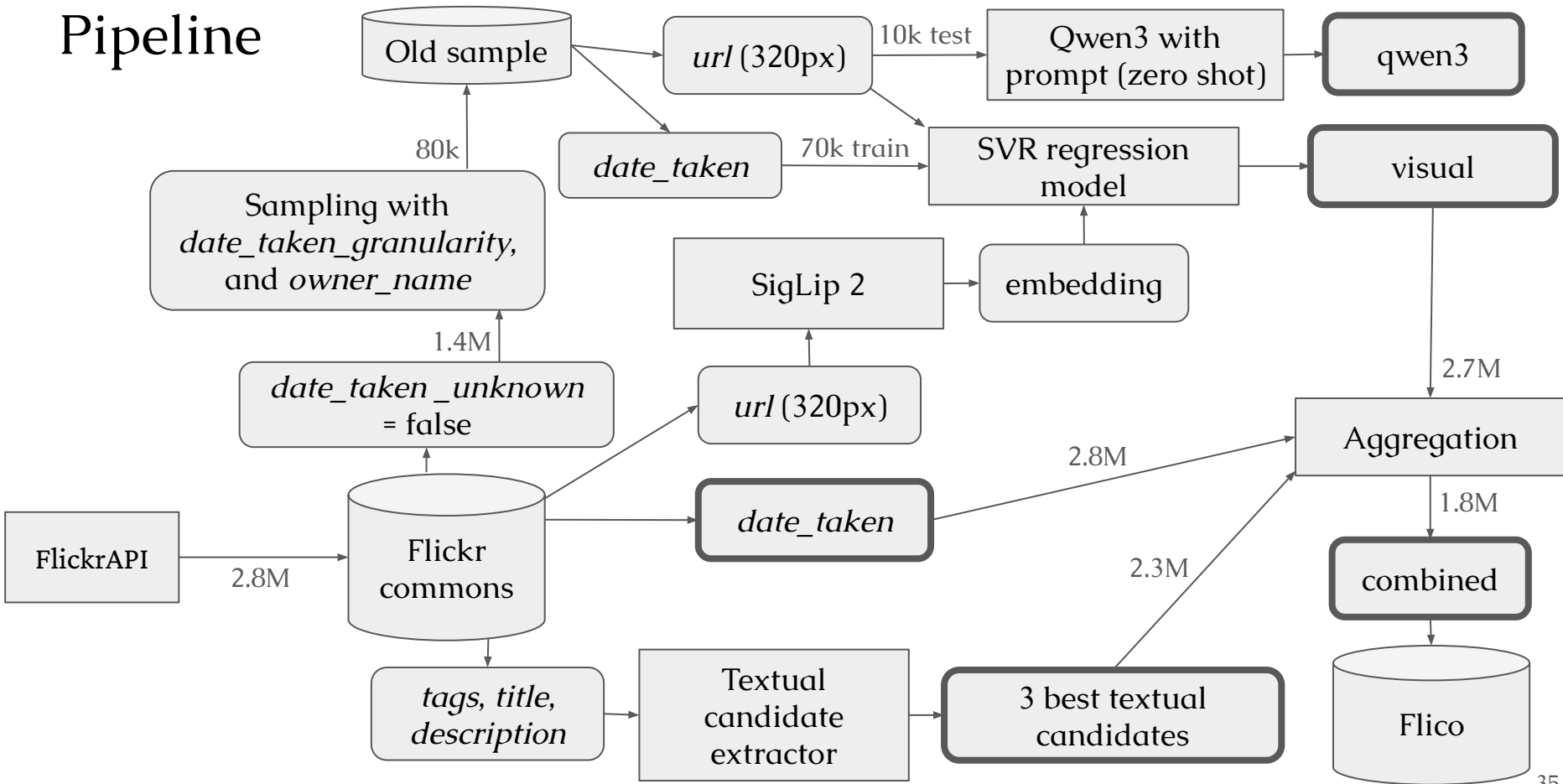








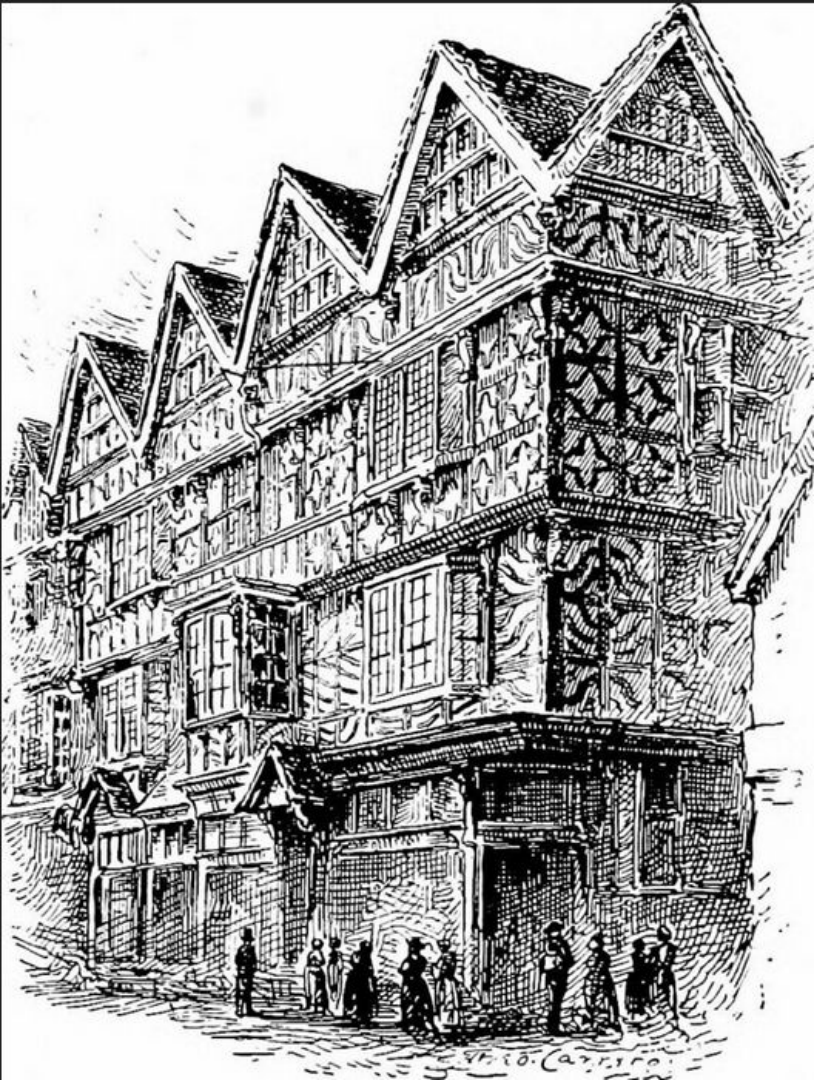
Pipeline





visual : 1922
qwen3 : None
date_taken : 1913
combined year: 1913

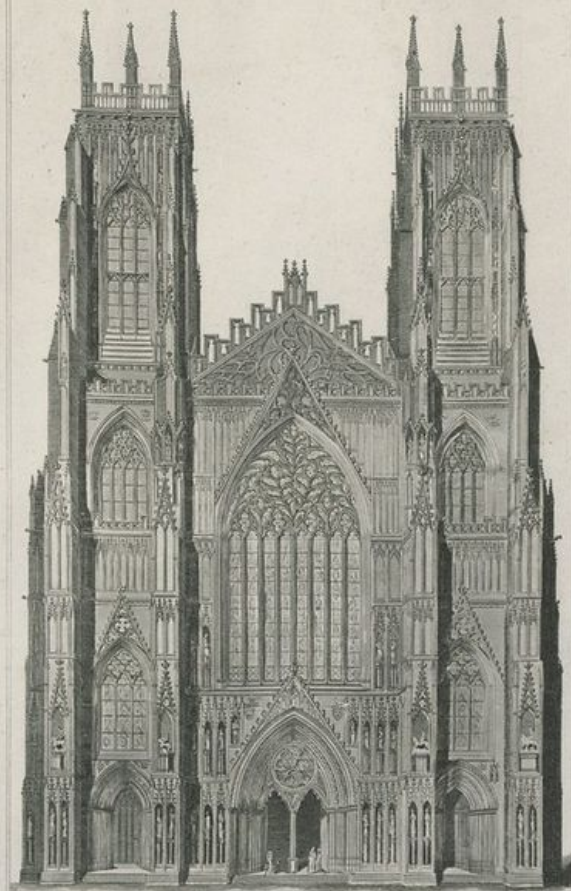
textual 1: 1913 p=0.801
textual 2: 1911 p=0.753
textual 3: 1912 p=0.722



visual : 1883
qwen3 : None
date_taken : 1908
combined_year: 1908

textual 1: 1908 p=1.000
textual 2: 1825 p=0.723
textual 3: 1914 p=0.722

<https://www.flickr.com/photos/126377022@N07/20135913734>



View of the
WEST FRONT OF YORK CATHEDRAL.
Published by J. B. Whittell & Co.

visual : 1915
qwen3 : None
date_taken : 2020
combined year: 1809

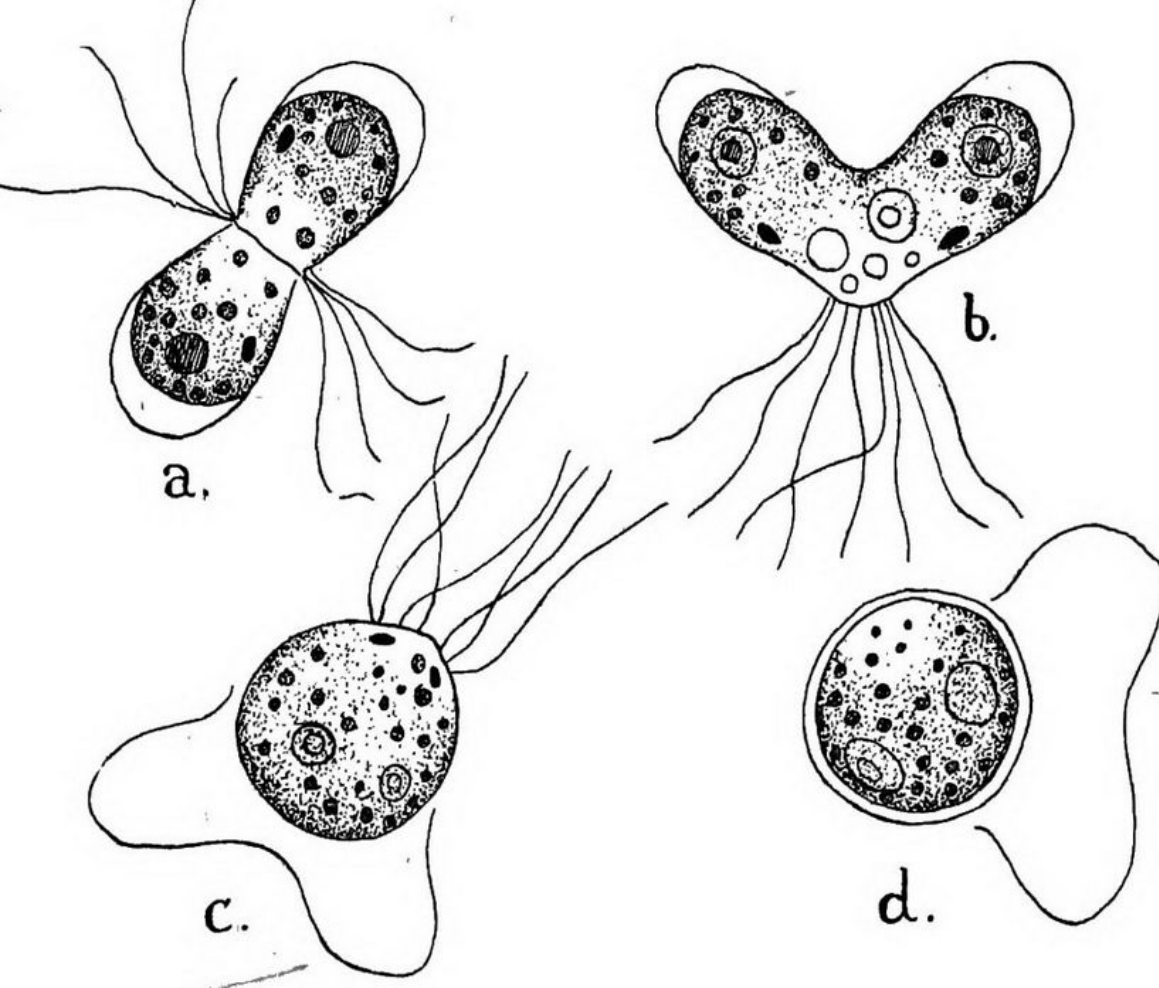
textual 1: 1809 p=1.000
textual 2: 1820 p=0.652
textual 3: 1738 p=0.638

<https://www.flickr.com/photos/12403504@N02/50264103686>



visual : 1961
qwen3 : None
date_taken : 2017
combined_year: 1957

textual 1: 1957 p=0.734
textual 2: p=0.000
textual 3: p=0.000



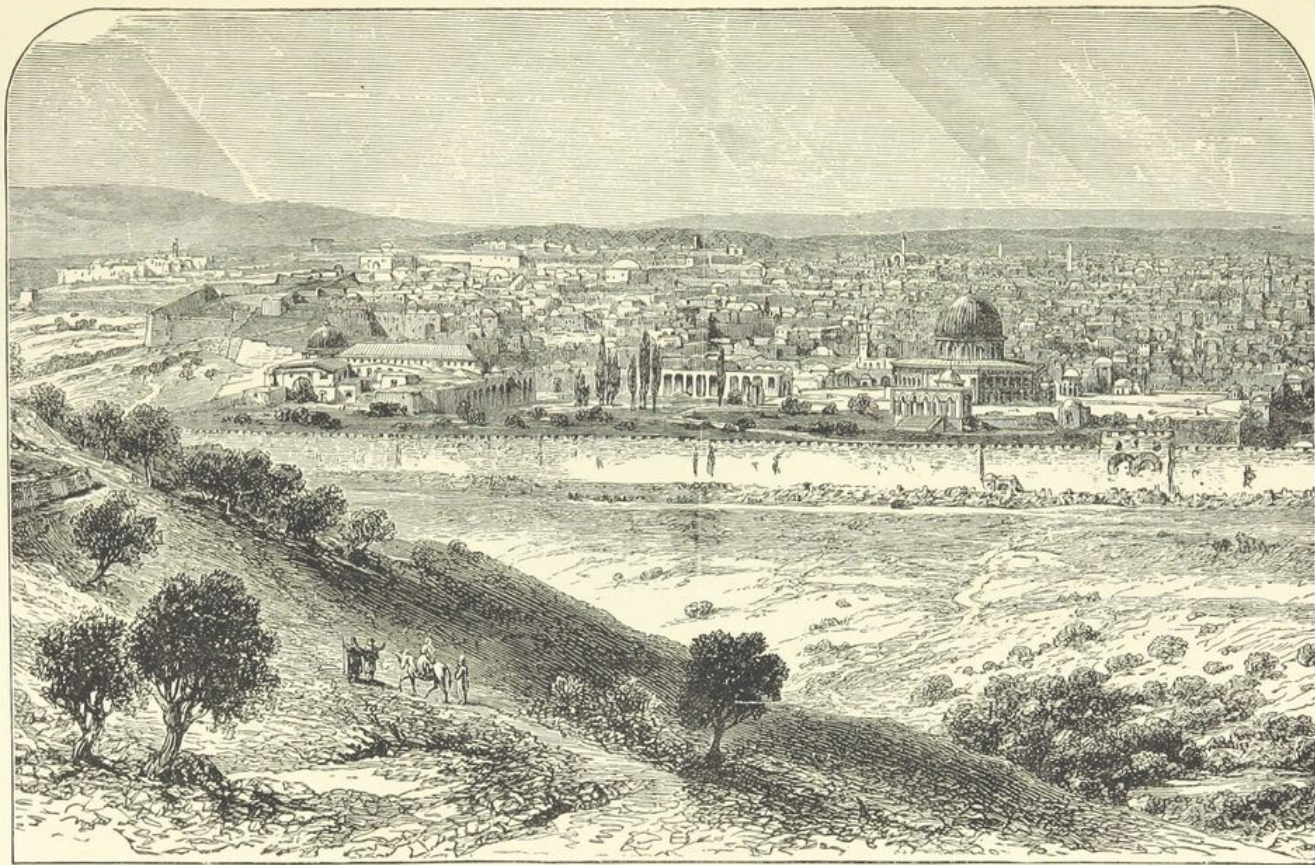
visual : 1929
qwen3 : None
date_taken : 1922
combined year: 1922

textual 1: 1922 p=1.000
textual 2: 1920 p=0.717
textual 3: 1871 p=0.713



visual : 1911
qwen3 : None
date_taken : 2024
combined year: 1925

textual 1: 1925 p=0.770
textual 2: p=0.000
textual 3: p=0.000



JERUSALEM FROM THE MOUNT OF OLIVES.

visual : 1896
qwen3 : None
date_taken : 2013
combined_year: 1893

textual 1: 1893 p=1.000
textual 2: p=0.000
textual 3: p=0.000



visual : 1961
qwen3 : None
date_taken : 2025
combined year: 1974

textual 1: 1974 p=0.776
textual 2: p=0.000
textual 3: p=0.000

<https://www.flickr.com/photos/49487266@N07/55006527278>

2693-5

JUDGE O. P. THOMPSON

H. T. RAINEY



visual : 1917
qwen3 : None
date_taken : 1910
combined_year: 1910

textual 1: 1915 p=0.853
textual 2: 2009 p=0.789
textual 3: 1910 p=0.716



visual : 1927
qwen3 : None
date_taken : 1915
combined year: 1915

textual 1: 1918 p=0.722
textual 2: 1914 p=0.722
textual 3: 1910 p=0.591



visual : 1897
qwen3 : None
date_taken : 1863
combined year: 1863

textual 1: 1865 p=0.930
textual 2: 1860 p=0.881
textual 3: 1861 p=0.740

Thank you !

<https://github.com/nathanaelambert/flico/>

<https://zenodo.org/records/20836373>